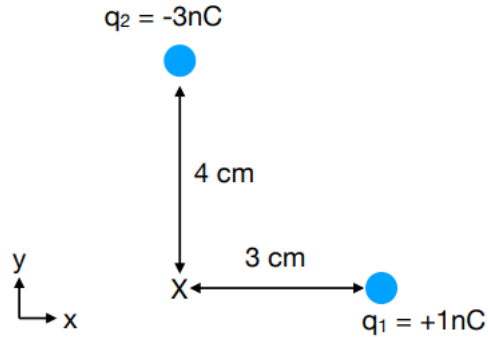


Electricity & Magnetism Homework 1

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Question 1

For the arrangement of charges shown in the diagram, calculate:



1. the Coulomb force between the two charges,
2. the electric field at point X,
3. the force on a 3rd charge of -2nC placed at point X.

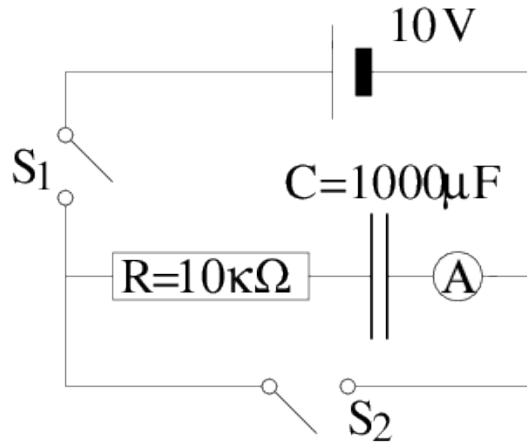
Question 2

Draw the electric field lines of a system consisting of:

1. two charges of equal magnitude and the same sign,
2. two charges of equal magnitude but the opposite sign, and
3. two charges, one with charge -5 C and one with charge $+3\text{ C}$.

Question 3

Look at the circuit below. The switch S_1 is closed and the capacitor is uncharged.



- a) Calculate the current measured in the ammeter immediately after switch S_1 is closed.
2. Calculate the current measured in the ammeter 10 s after closing switch S_1 .
3. Calculate the current measured in the ammeter 5 minutes after closing switch S_1 .
4. After 10 minutes, the switch S_1 is open and switch S_2 is closed. Calculate the current in the ammeter immediately after S_2 is closed.
5. Calculate the current in the ammeter 10 s after S_2 is closed.
6. Calculate the current in the ammeter 5 minutes after S_2 is closed.